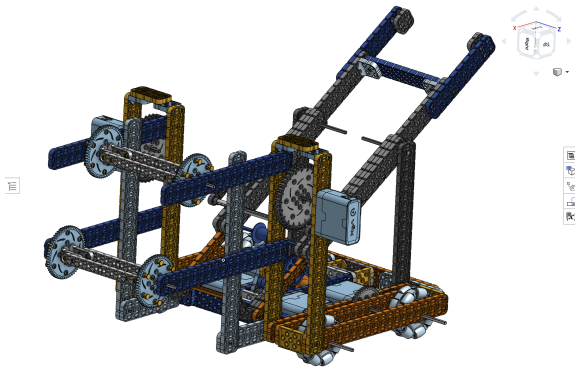
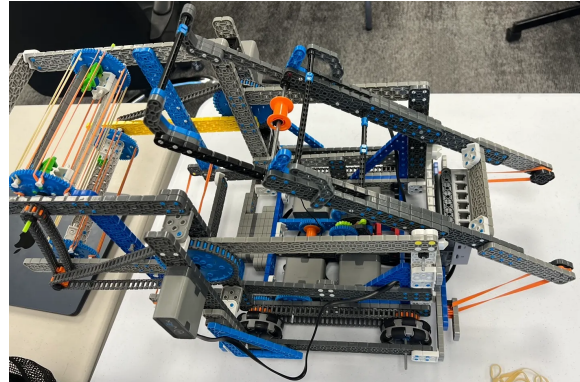


Lesson 21-22: Physically Building a Competition Robot for Rapid Relay



The 3D model we build from the last lesson



The actual robot from the 3D model

In this lesson, you'll turn your 3D model into a tangible, working competition robot for the Rapid Relay challenge! Utilizing VEX IQ components, you'll bring your design to life and test it under real-world conditions to see how it performs.

Instructions:

1. Review Your 3D Model:

- Start by revisiting your completed 3D model of the competition robot from the previous lesson. Pay special attention to the drivetrain, arm mechanisms, and any gear systems you've designed. Ensure you understand how each part contributes to your robot's performance in the Rapid Relay challenge.
- Take note of any areas where you expect potential stress or movement limitations. Think about how you'll ensure stability and accuracy in your physical build.

2. Gather VEX IQ Components:

- Collect all the VEX IQ parts you'll need based on your 3D model. This includes motors, gears, wheels, plates, and any sensors or specialty

pieces you've planned to use.

- Make sure you have enough parts to fully replicate your digital design.

3. Build the Robot's Drivetrain:

- Begin construction by assembling the drivetrain, using your 3D model as a guide. Make sure the structure is solid, and all axles and gears rotate smoothly. Test its movement as you go to avoid any alignment issues.
- Verify that the motors are securely connected to the drivetrain and are positioned according to your design.

4. Construct the Arm Mechanisms:

- Follow your model to build any moving arms or lifting mechanisms required for the competition. Make sure they are sturdy and that the gear ratios provide the desired speed and torque.
- Test the functionality of these mechanisms, ensuring they can handle the load they'll encounter in Rapid Relay.

5. Test the Robot's Functionality:

- Power up your robot and run through a few simple tests to confirm that everything is functioning correctly. If any components aren't working as expected, compare them with your 3D model to identify discrepancies.
- Practice driving the robot, lifting objects, and completing Rapid Relay tasks. Observe its stability, movement, and how it handles obstacles.

6. Refine and Adjust:

- If you encounter any problems, make necessary adjustments. This is the stage where your 3D design may need refinement based on real-world testing.
- Ensure all parts are securely fastened and that your robot is performing as you intended.

By the end of this lesson, you will have a competition-ready robot for Rapid Relay, designed from scratch and built to match your vision. This hands-on experience will solidify your understanding of the design-build-test cycle and showcase how your 3D model transforms into a functioning robot.